

PART 1: INTRODUCTION TO SIEM

Research and explain:

- 1. What is SIEM?**
- 2. Why do organizations use SIEM?**
- 3. Difference between SIEM and Log Management.**
- 4. Popular SIEM Tools:**
 - **Splunk**
 - **IBM QRadar**
 - **Microsoft Sentinel**
 - **ELK Stack**

PART 2: SPLUNK INSTALLATION

- 1. Install Splunk Enterprise (Free Trial).**
- 2. Launch Splunk.**
- 3. Explore the following sections:**
 - **Search & Reporting**
 - **Dashboards**
 - **Alerts**
 - **Reports**

Take screenshots.

PART 3: DATA INGESTION

Upload any sample log file into Splunk.

Examples:

- **Web Server Logs**
- **Windows Logs**
- **Application Logs**

Tasks:

- 1. Import the log file.**
- 2. Verify data ingestion.**
- 3. Count total events.**

Take screenshots.

PART 4: SEARCH & ANALYSIS

Perform the following searches:

- 1. View all events.**
- 2. Find failed login events.**
- 3. Identify top source IP addresses.**
- 4. Count unique IP addresses.**
- 5. Find events with HTTP 404 errors.**

Document your observations.

PART 5: DASHBOARD CREATION

Create a dashboard showing:

- Total Events**
- Top IP Addresses**
- Top Error Codes**
- Event Distribution**

Take screenshots.

ASSIGNMENT

Prepare a report containing:

- 1. Introduction to SIEM**
- 2. Introduction to Splunk**
- 3. Screenshots of Installation**
- 4. Data Ingestion Process**
- 5. Search Queries Used**
- 6. Dashboard Screenshots**
- 7. Findings and Observations**

Create an alert for:

"More than 5 failed login attempts from the same IP."

Explain:

- Why is this suspicious?**
- What actions should a SOC Analyst take?**

SUBMISSION

- PDF or Word Document**

PART 1: INTRODUCTION TO SIEM

What is SIEM?

Security Information and Event Management (SIEM) is a cybersecurity solution that collects, stores, analyzes, and monitors logs from various systems and applications. It helps organizations detect suspicious activities and security threats in real time. SIEM combines security event management with log management to provide centralized monitoring.

Why do organizations use SIEM?

Organizations use SIEM to improve their security monitoring and incident response capabilities. It helps in detecting attacks early, investigating suspicious behavior, and maintaining compliance with security regulations. SIEM also automates alerts, reducing manual work.

Purpose	Description
Threat Detection	Detects malicious activities quickly
Centralized Monitoring	Monitors logs from all devices in one place
Compliance	Helps meet legal requirements
Faster Response	Reduces response time during incidents

Difference between SIEM and Log Management

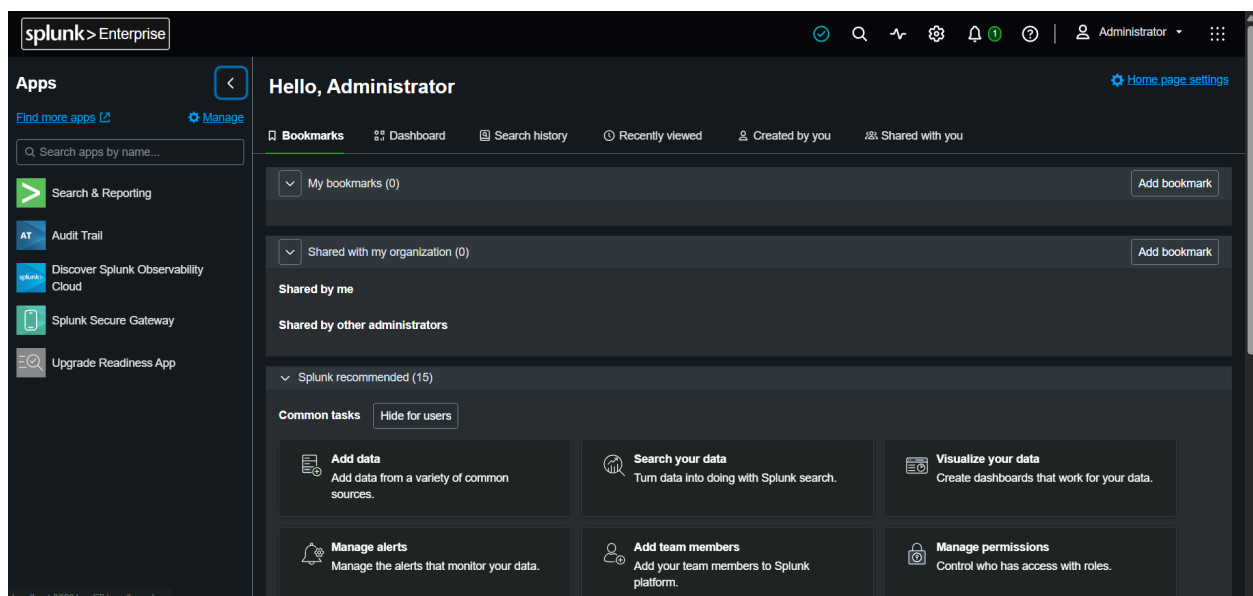
SIEM	Log Management
Provides security monitoring and analysis	Only stores logs
Generates alerts for threats	No threat detection
Helps investigate attacks	Limited analysis
Supports incident response	Mainly for storage

Popular SIEM Tools

Tool	Description
Splunk	Advanced log analysis and monitoring
IBM QRadar	Enterprise security monitoring tool
Microsoft Sentinel	Cloud-based SIEM platform
ELK Stack	Open-source log management tool

PART 2: INTRODUCTION TO SPLUNK

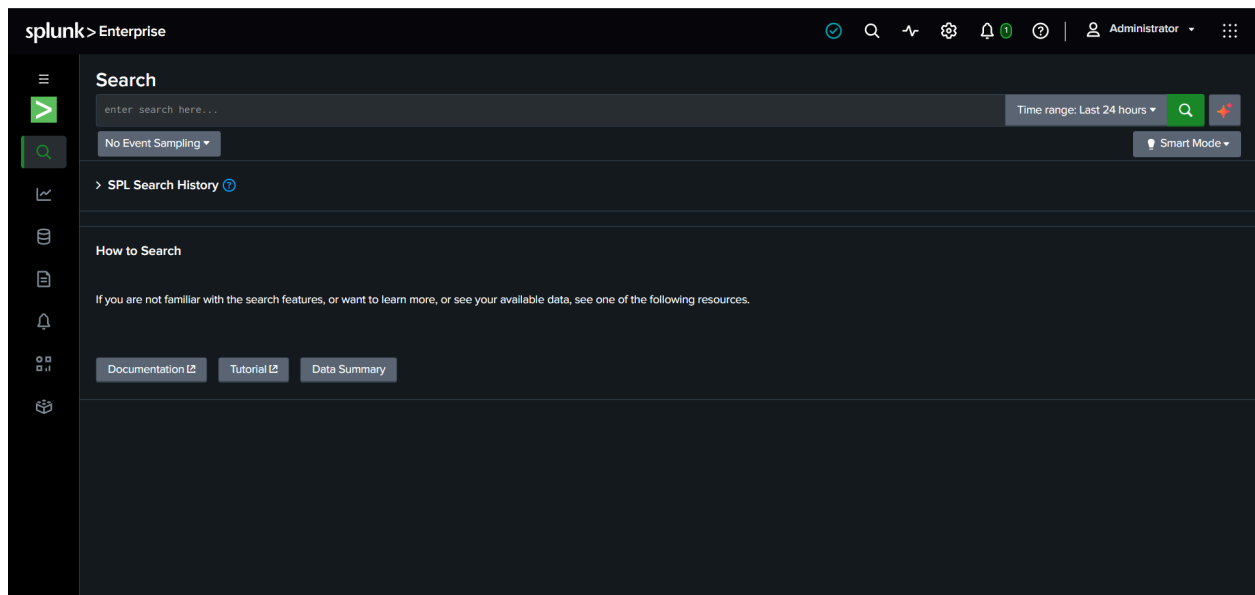
Splunk Enterprise is a powerful SIEM platform used to collect, index, search, and visualize machine-generated data. It helps security teams monitor systems and detect threats quickly.



Splunk provides several important sections:

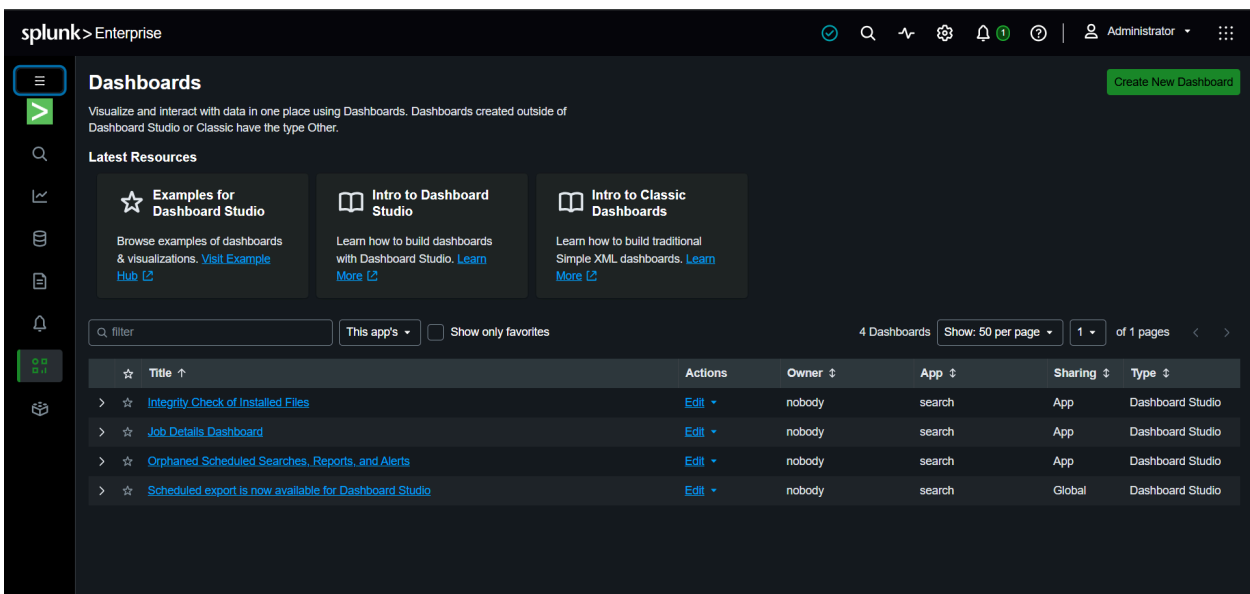
Search & Reporting

This section is used to search logs using SPL (Search Processing Language). Analysts can filter logs, detect suspicious events, and create reports.



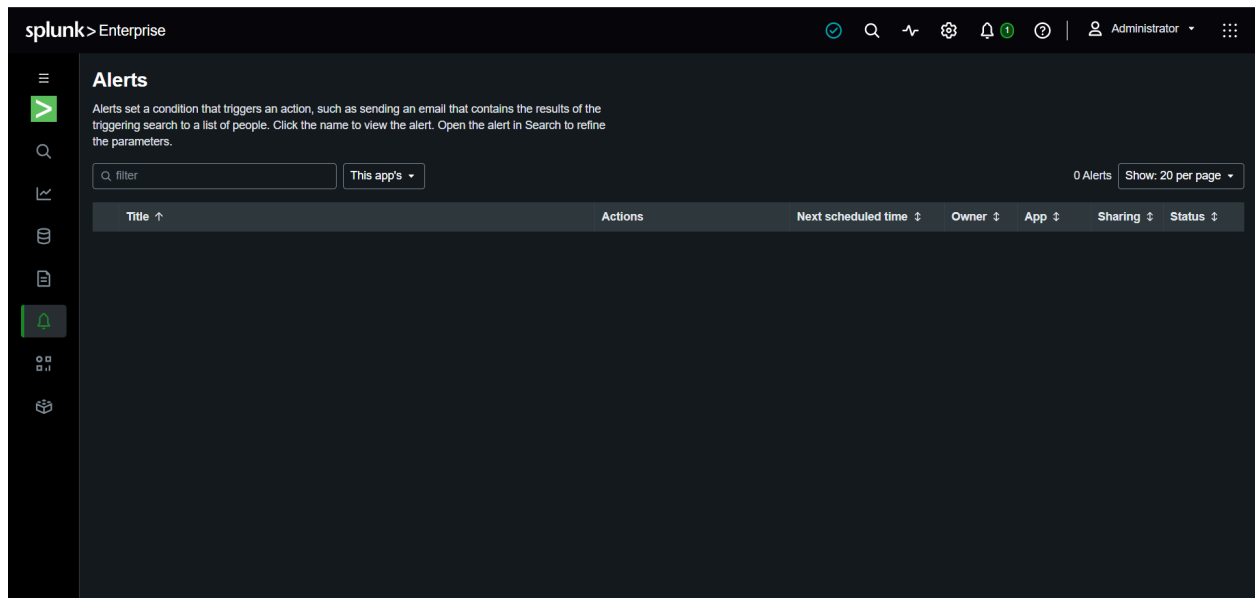
Dashboards

Dashboards provide graphical representation of logs and events. They help analysts understand data quickly using charts, graphs, and statistics.



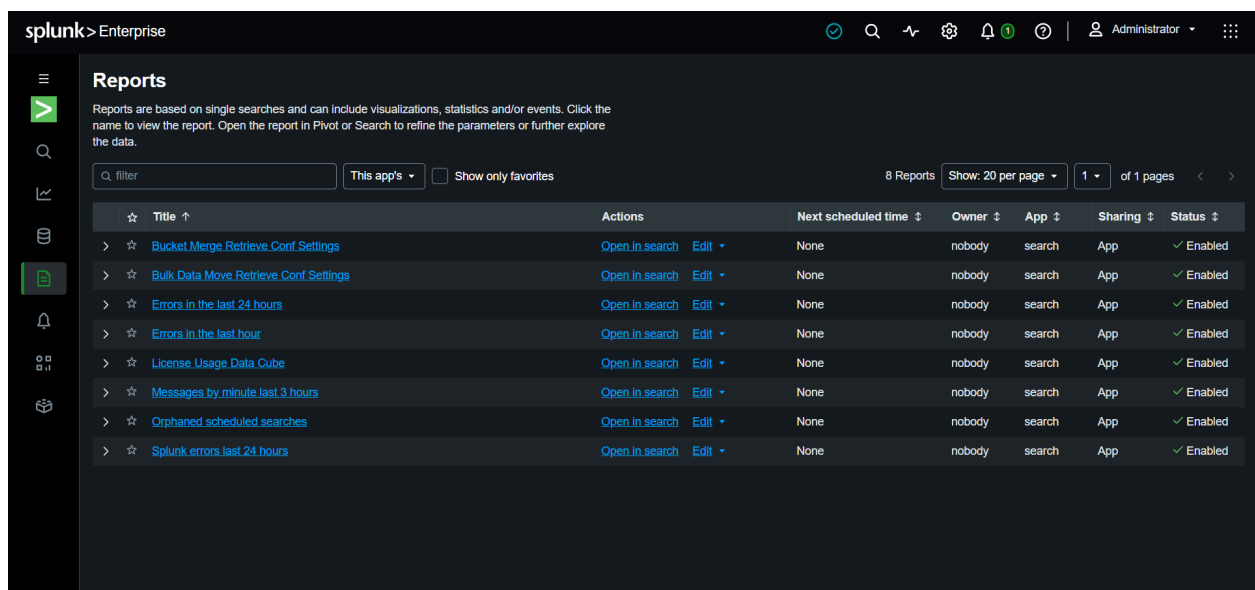
Alerts

Alerts are automated notifications triggered when specific conditions are met. They help analysts detect attacks like brute-force attempts without manually checking logs.



Reports

Reports are saved search results that can be used later for analysis. They help in documentation and regular security monitoring.



PART 3: DATA INGESTION

Data ingestion means importing logs into Splunk so they can be analyzed. Splunk supports many log types such as web logs, system logs, and application logs.

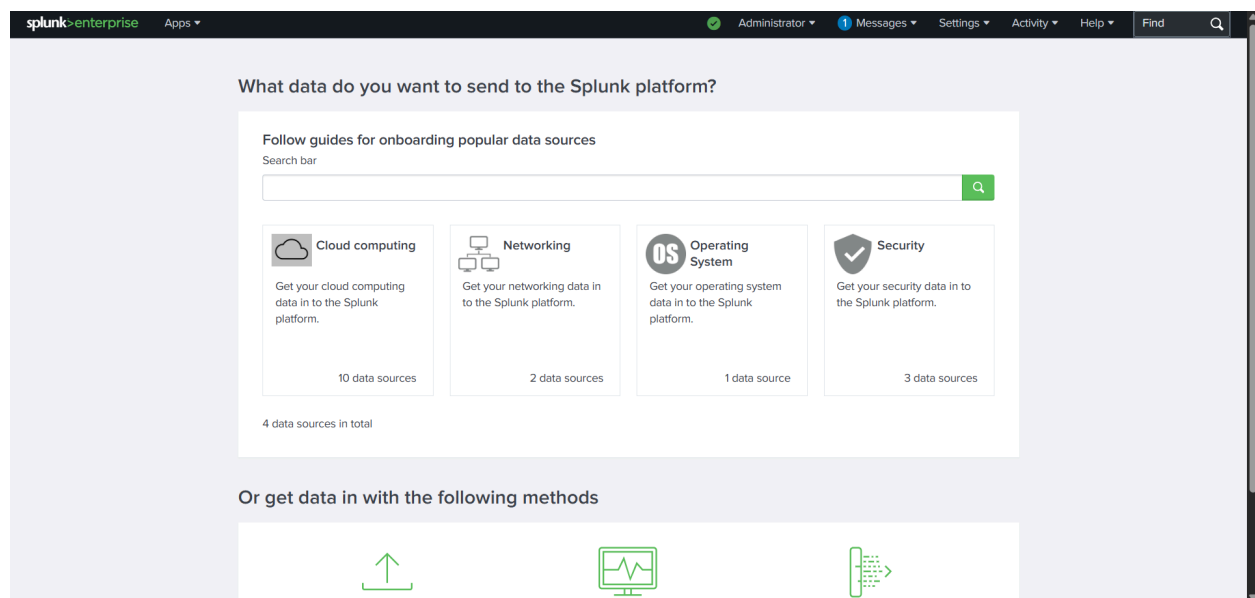
In this practical, sample log files were uploaded.

Steps performed:

1. Open Add Data
2. Upload log file
3. Select source type
4. Review settings
5. Submit data

Upload Log File

This step allows importing external log files into Splunk.



Source Type Selection

Source type tells Splunk what kind of log is being uploaded so it can parse the data correctly.

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **access.log** [View Event Summary](#)

Source type: **Select Source Type** **Save As**

	Time	Event
1	25/06/2026 10:00:01.000	192.168.1.10 - - [25/Jun/2026:10:00:01] "GET /index.html HTTP/1.1" 200 1024 "Mozilla/5.0"
2	25/06/2026 10:01:15.000	192.168.1.11 - - [25/Jun/2026:10:01:15] "POST /login HTTP/1.1" 401 512 "Chrome/120"
3	25/06/2026 10:02:30.000	192.168.1.12 - - [25/Jun/2026:10:02:30] "GET /admin HTTP/1.1" 404 256 "curl/7.68"
4	25/06/2026 10:03:10.000	192.168.1.10 - - [25/Jun/2026:10:03:10] "GET /index.html HTTP/1.1" 200 1024 "Mozilla/5.0"
5	25/06/2026 10:04:20.000	192.168.1.11 - - [25/Jun/2026:10:04:20] "POST /login HTTP/1.1" 401 512 "Chrome/120"

Review Settings

Before final upload, Splunk shows all selected settings for confirmation.

Review

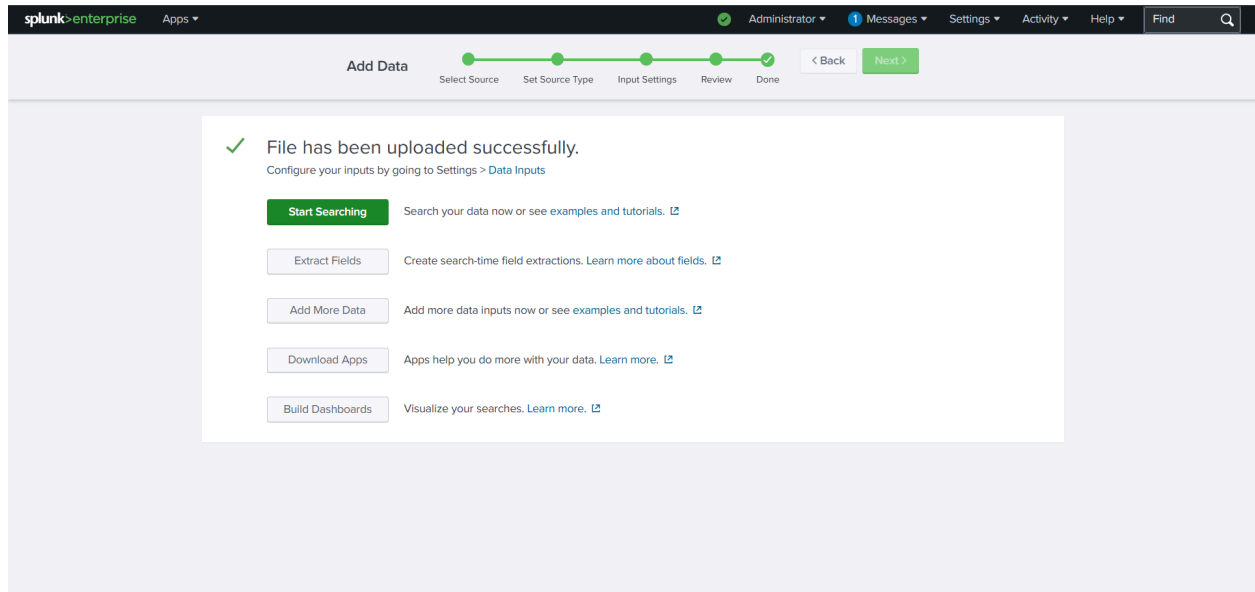
Input Type Uploaded File
File Name access.log
Source Type apache_access
Host TUSHAR
Index main

Submit

localhost:8000/en-GB/manager/search/adddatamethods/review#

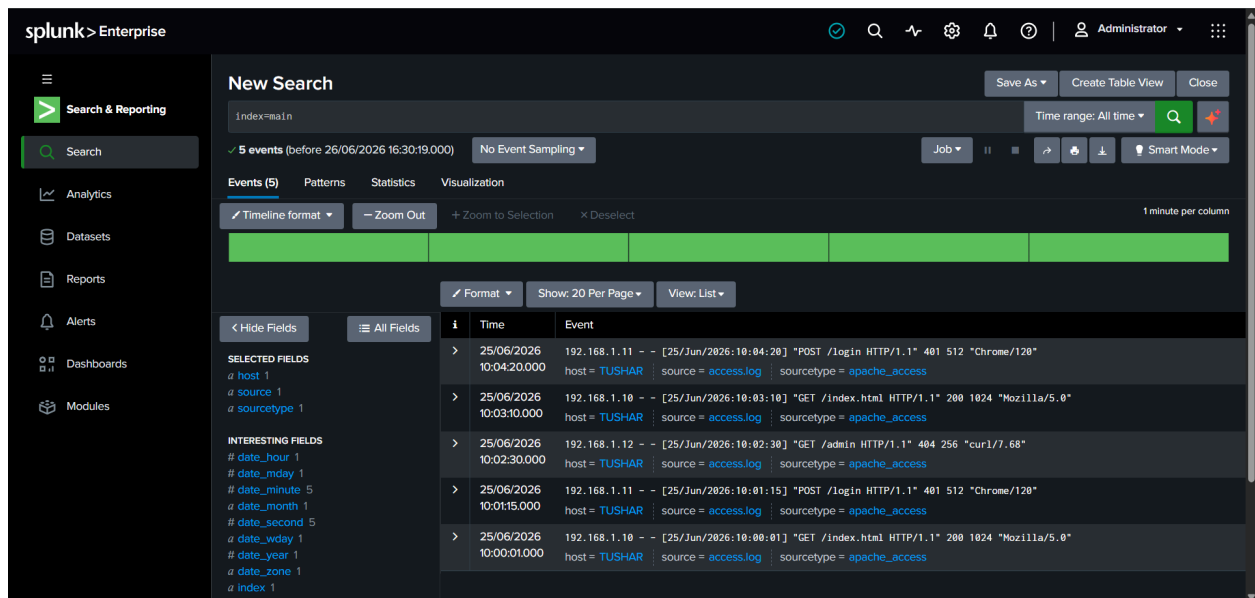
Successful Data Ingestion

After submission, logs are indexed and ready for searching.



Total Event Count

This shows the total number of events imported into Splunk.



PART 4: SEARCH AND ANALYSIS

Splunk uses SPL queries for searching logs. Different queries were used for analysis.

View All Events

Query:
index=*

This query displays all available events from the uploaded logs.

The screenshot shows the Splunk Enterprise web interface. The top navigation bar includes the Splunk logo, user profile (Administrator), and various icons. The left sidebar contains a 'Search & Reporting' section with a 'Search' button and other options like Analytics, Datasets, Reports, Alerts, Dashboards, and Modules. The main content area is titled 'New Search' and displays the query 'index=*main'. Below the query, it shows '5 events (before 26/06/2026 16:30:19.000)' and a 'No Event Sampling' dropdown. The search results are displayed in a table with columns for Time and Event details. The table shows five events, each with a timestamp, host, source, and sourcetype.

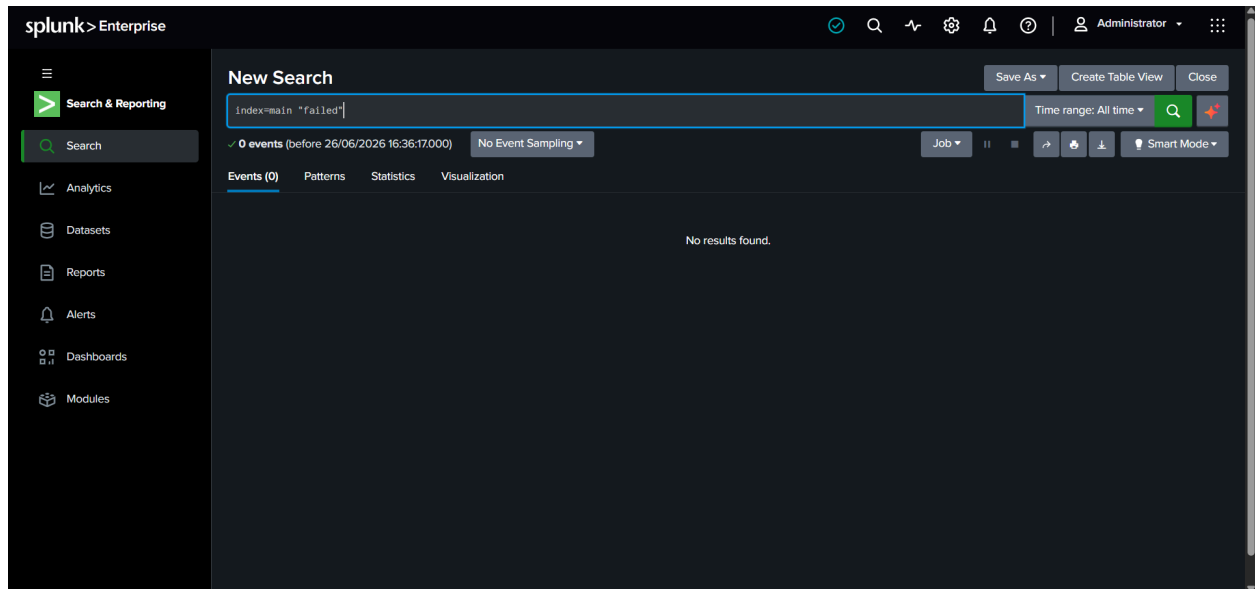
i	Time	Event
>	25/06/2026 10:04:20.000	192.168.1.11 - [25/Jun/2026:10:04:20] "POST /login HTTP/1.1" 401 512 "Chrome/120" host = TUSHAR source = access.log sourcetype = apache_access
>	25/06/2026 10:03:10.000	192.168.1.10 - [25/Jun/2026:10:03:10] "GET /index.html HTTP/1.1" 200 1024 "Mozilla/5.0" host = TUSHAR source = access.log sourcetype = apache_access
>	25/06/2026 10:02:30.000	192.168.1.12 - [25/Jun/2026:10:02:30] "GET /admin HTTP/1.1" 404 256 "curl/7.68" host = TUSHAR source = access.log sourcetype = apache_access
>	25/06/2026 10:01:15.000	192.168.1.11 - [25/Jun/2026:10:01:15] "POST /login HTTP/1.1" 401 512 "Chrome/120" host = TUSHAR source = access.log sourcetype = apache_access
>	25/06/2026 10:00:01.000	192.168.1.10 - [25/Jun/2026:10:00:01] "GET /index.html HTTP/1.1" 200 1024 "Mozilla/5.0" host = TUSHAR source = access.log sourcetype = apache_access

Observation:
All logs were visible successfully.

Find Failed Login Events

Query:
index=* "failed"

This query filters logs related to failed authentication attempts.

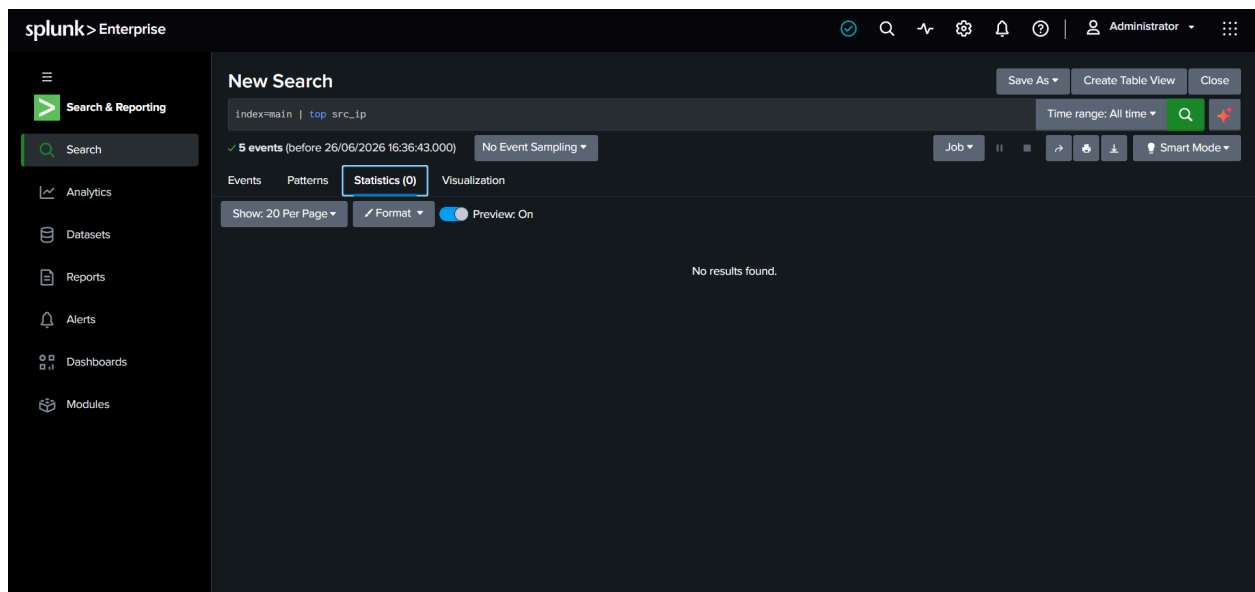


Top Source IP Addresses

Query:

`index=* | top src_ip`

This query identifies the IP addresses generating the highest number of events.

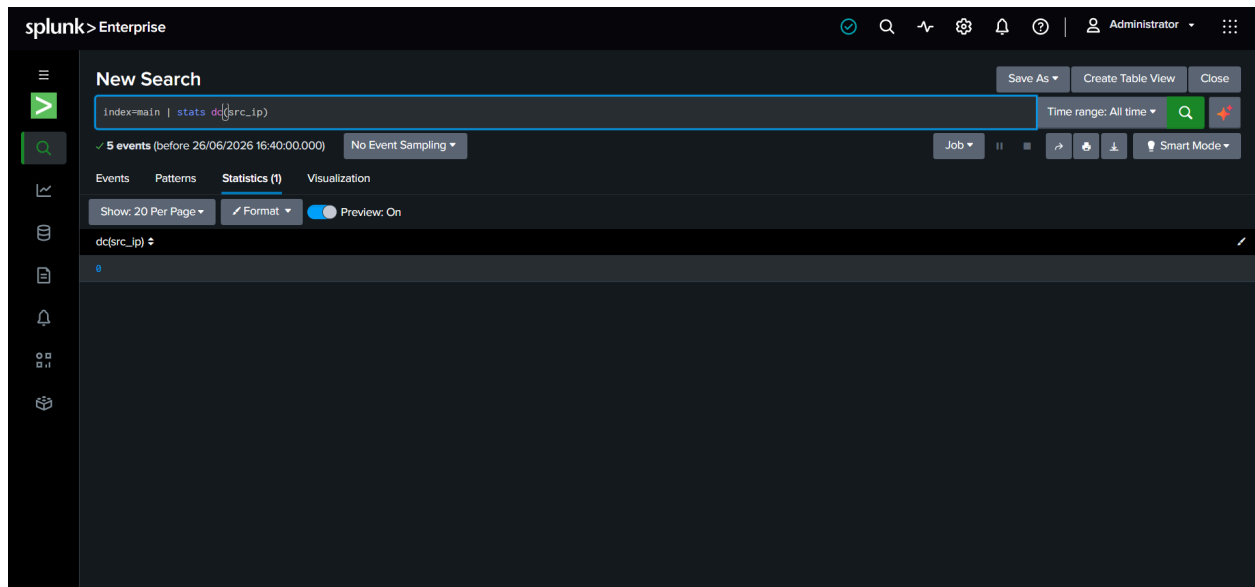


Count Unique IP Addresses

Query:

```
index=* | stats dc(src_ip)
```

This query counts distinct IP addresses in the logs.



Observation:

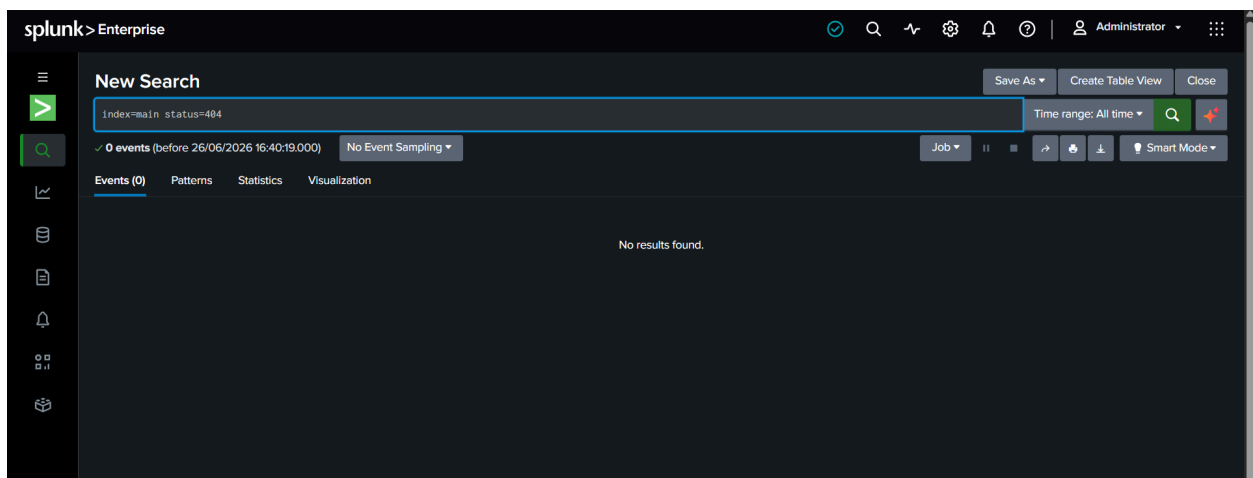
Unique IP count was calculated.

Find HTTP 404 Errors

Query:

```
index=* status=404
```

This query detects failed webpage requests resulting in 404 errors.

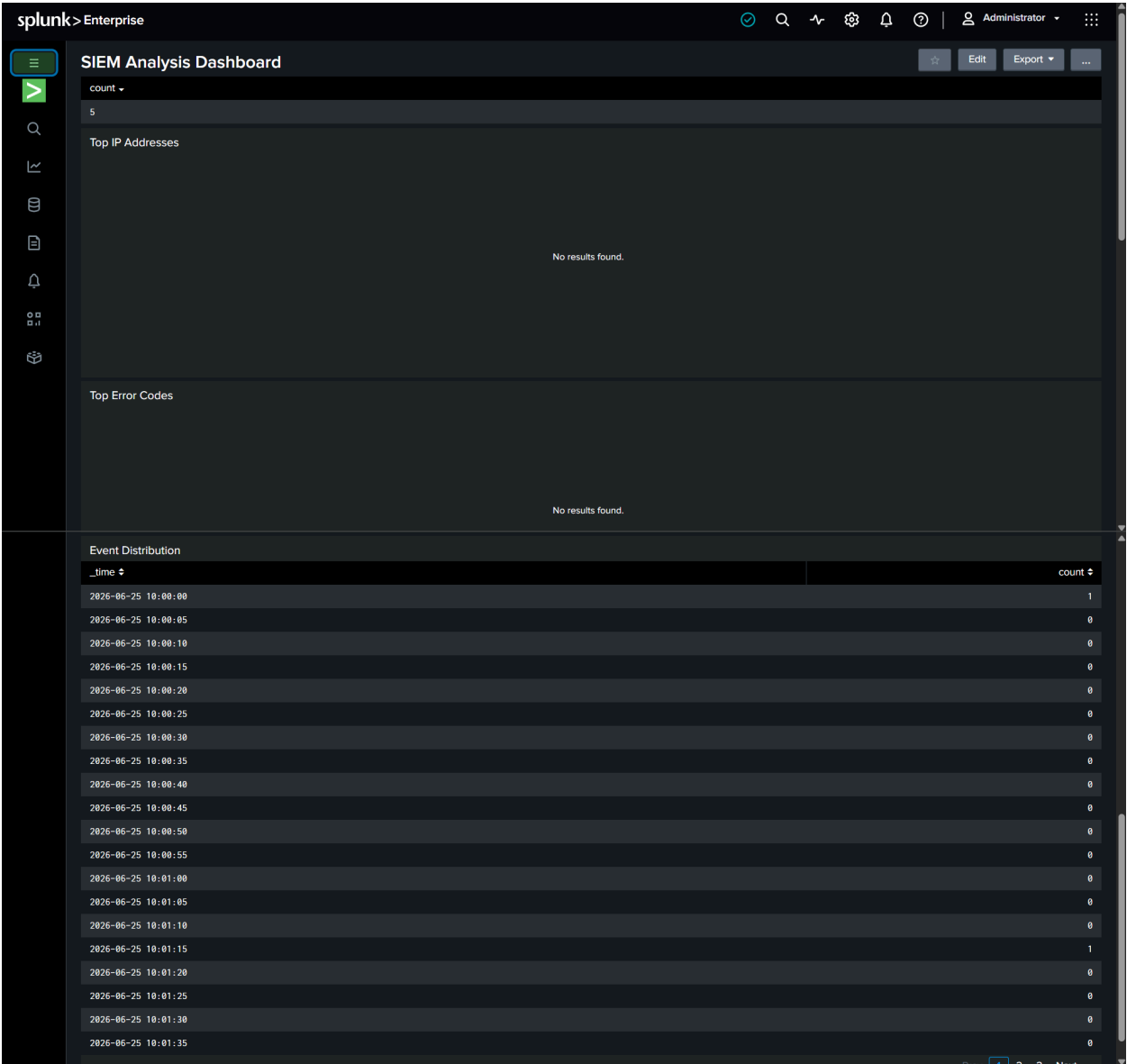


PART 5: DASHBOARD CREATION

Dashboards are useful for visualizing log data in a structured format. They allow analysts to monitor key metrics without running queries repeatedly.

A dashboard was created showing:

Dashboard Panel	Purpose
Total Events	Displays total log count
Top IP Addresses	Shows most active IPs
Top Error Codes	Displays common errors
Event Distribution	Shows pattern of events



Observation:

Dashboard made data easier to understand visually.

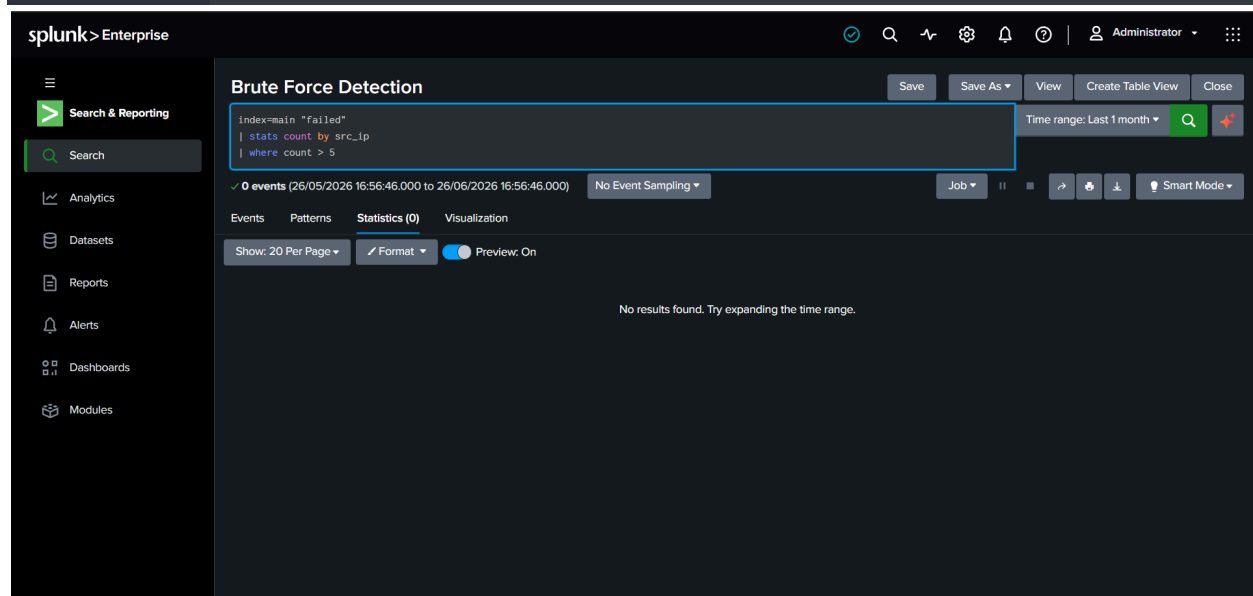
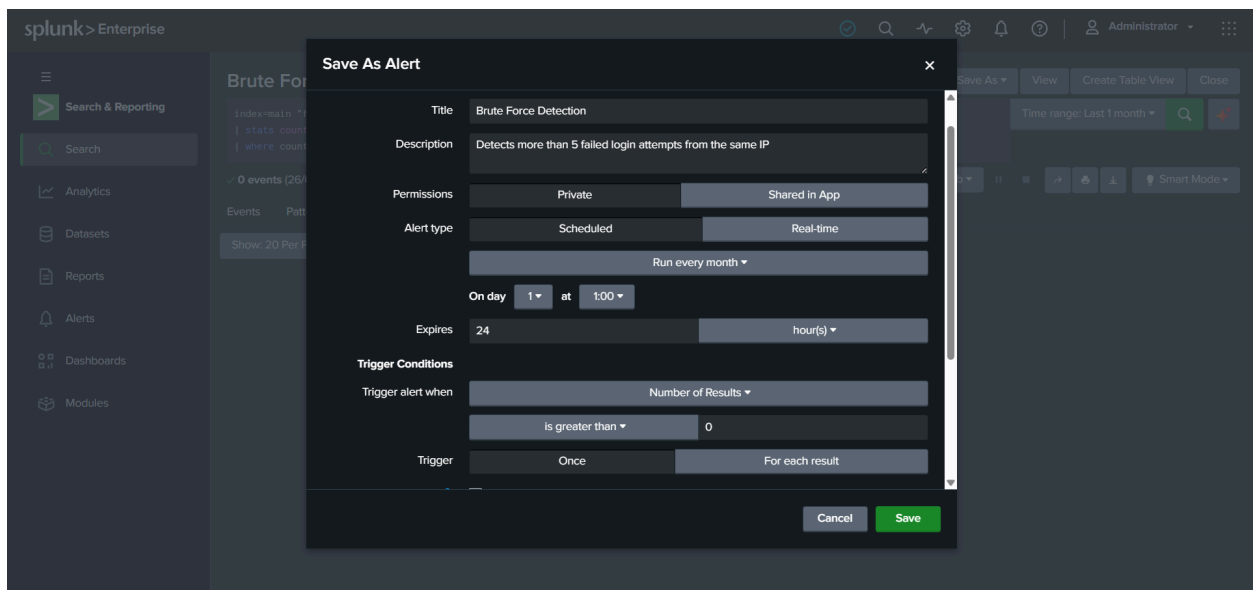
PART 6: ALERT CREATION

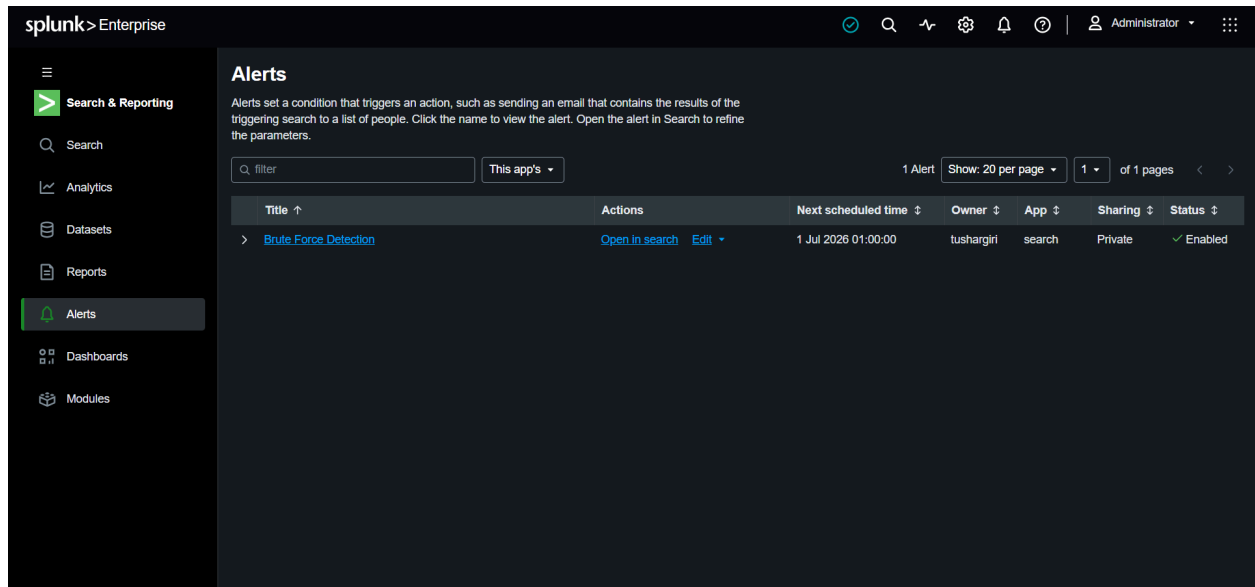
Alerts help automate threat detection. Instead of checking logs manually, Splunk triggers alerts when suspicious activity is detected.

Query:

```
index=* "Failed password"  
| stats count by src_ip  
| where count > 5
```

This alert triggers when an IP attempts more than 5 failed logins.





Why is this suspicious?

Multiple failed login attempts from the same IP can indicate a brute-force attack. Attackers try many passwords to gain unauthorized access. This can compromise accounts and sensitive information.

SOC Analyst Actions

Action	Description
Investigate IP	Check if the IP is malicious
Review Logs	Analyze attack details
Block IP	Stop future attempts
Reset Passwords	Secure user accounts
Notify Team	Escalate incident if necessary

FINDINGS AND OBSERVATIONS

- SIEM helps centralize and monitor logs.
- Splunk successfully ingested log files.
- Search queries helped identify failed logins and errors.
- Top source IPs were detected.
- Dashboard visualization improved understanding.
- Alerts automated brute-force detection.

CONCLUSION

Splunk Enterprise is an effective SIEM solution for security monitoring, log analysis, and threat detection. Through this practical, it was observed that Splunk simplifies log management and helps detect suspicious activities like brute-force attacks efficiently.